

Solution – Travelling Salesman Problem

Consider 4 cities: **A, B, C, D**

Given distances:

- A to B = 10
- A to C = 15
- A to D = 20
- B to C = 35
- B to D = 25
- C to D = 30

Steps

1. Start from city **A**.
2. Possible routes are:
 - $A \rightarrow B \rightarrow C \rightarrow D \rightarrow A = 10 + 35 + 30 + 20 = \mathbf{95}$
 - $A \rightarrow B \rightarrow D \rightarrow C \rightarrow A = 10 + 25 + 30 + 15 = \mathbf{80}$
 - $A \rightarrow C \rightarrow B \rightarrow D \rightarrow A = 15 + 35 + 25 + 20 = \mathbf{95}$
 - $A \rightarrow C \rightarrow D \rightarrow B \rightarrow A = 15 + 30 + 25 + 10 = \mathbf{80}$
 - $A \rightarrow D \rightarrow B \rightarrow C \rightarrow A = 20 + 25 + 35 + 15 = \mathbf{95}$
 - $A \rightarrow D \rightarrow C \rightarrow B \rightarrow A = 20 + 30 + 35 + 10 = \mathbf{95}$
3. Compare the total distances.
4. The minimum distance is **80 units**.
5. Therefore, the optimal route is **$A \rightarrow B \rightarrow D \rightarrow C \rightarrow A$** .

Result

Optimal Route: $A \rightarrow B \rightarrow D \rightarrow C \rightarrow A$

Minimum Total Distance: **80 units** 